

Aneesh Krishna

tinyurl.com/aneesh495 | linkedin.com/in/aneesh495 | github.com/aneesh495

EDUCATION

Purdue University

May 2027

BS Computer Science & Statistics

Minors: Math & Finance

- **Courses:** Computer Architecture · Systems Programming · Analysis of Algorithms
- **Orgs/Awards:** USACO Silver · 6x Hackathon Winner · Boiler Blockchain · Data Mine

EXPERIENCE

Amazon | *Systems Development Intern*

May 2026 – Aug 2026

- Built a FastAPI automation service on AWS, cutting tier-generation time by 98% across 100MiB Excel workbooks
- Orchestrated a 6-stage async data processing pipeline, scaling backend throughput by 20x across 10,000+ jobs
- Developed a low-level XML ZIP manipulation layer, ensuring 100% byte-fidelity for 5,000 corporate assets
- Automated live distributed data enrichment from Redshift, eliminating 12,000 manual reconciliation steps
- Secured critical pricing endpoints using SSO, managing zero-trust RS256 token validation for 800 internal users

Caterpillar | *Machine Learning Intern*

Aug 2025 – Dec 2025

- Architected a PyTorch multi-horizon transformer, boosting supply-chain forecast accuracy by 24% across 1M+ records
- Scaled distributed ML inference via Kubernetes, slashing forecasting latency for 500+ SKUs down to 2.85 minutes
- Deployed an automated Azure MLOps pipeline, processing 100K+ daily telemetry streams for real-time model serving

Playtoon It | *Software Developer*

Jun 2025 – Aug 2025

- Optimized ECS & Netcode to lower server costs by 30% and packet loss by 25% (30,000 player sessions)
- Cut latency spikes by 40% and player churn by 20% through event-driven architecture and object pooling

Retail Hedging | *Software Engineer*

Mar 2025 – May 2025

- Implemented Auth0–Stripe payment gateway with RS256 JWT verification, securing 20,000 checkout sessions
- Tripled API throughput by replacing high-frequency client polling with tokenized API flow, reducing load by 70%

PROJECTS

Cathode - CPU Graphics Engine

- Built a zero-dependency CPU graphics engine via AArch64 NEON SIMD, rendering 50+ scenes at sub-2ms latency
- Designed real-time NTSC signal DSP backed by a 115-point cross-language test suite across 24k lines of C/Rust

POSIX – *Custom UNIX Shell*

- Handles 30 core Bash features with process control & I/O redirection, lowering command dispatch latency by 60%
- Supports 500 concurrent jobs with sub-5ms scheduling via fork/exec/waitpid, cutting zombie overhead by 45%

Vibe – *Social Network*

- Scaled to 25,000 load-tested users with sub-50ms p95 latency using thread pooling and message queues
- Improved session reliability by 75% across 6,000 logins through persistent user authentication and session locks

SKILLS

- **Languages:** Python · TypeScript · C/C++ · Go · Rust · Java · SQL · R · C# · Swift · Kotlin · Ruby
- **Frameworks:** Next.js · FastAPI · LangChain · PyTorch · TensorFlow · Redis · Kafka · RabbitMQ · Valgrind
- **Tools:** Git · Docker · AWS · MongoDB · Firebase · Linux · Terraform · Kubernetes · Prometheus · Grafana · gRPC